1 If the Goal Is to Represent u, Why Do We Study the Flux $f(u)$?

1.1 1. Definitions

The quantity u is the physical quantity that we ultimately want to know and evolve.

However, for a conservation law, the partial differential equation (PDE) tells us that the value of u changes because the quantity is transported from one region to another through a **flux** $f(u)$.

Therefore,

$$\boxed{u = \text{quantity stored locally}}$$

while

$$\boxed{f(u) = \text{rate at which that quantity is transported through space}}$$

The flux does not replace u. Instead, it tells us **how u changes with time**.

1.2 2. Start from the Conservation Law

A one-dimensional conservation law has the general form

$$\boxed{\frac{\partial u}{\partial t} + \frac{\partial f(u)}{\partial x} = 0}$$

There are two main objects:

- $u(x,t)$: the physical quantity that we want to determine;
- $f(u)$: the flux, which describes how that quantity is transported through space.

The equation can also be written as

$$\frac{\partial u}{\partial t} = -\frac{\partial f(u)}{\partial x}.$$

Therefore,

$$\boxed{\text{time variation of } u = -\text{spatial variation of the flux}}$$

In words:

The amount of u at a particular location changes because the amount flowing into that region is different from the amount flowing out.

1.3 3. Why Can We Not Just Evolve u Directly?

The PDE does not tell us something arbitrary such as

$$u^{n+1} = u^n + \text{some arbitrary correction.}$$

Instead, it gives the evolution law

$$u_t = -f(u)_x.$$

Thus, the future state is determined by the behavior of the flux.

For a finite-volume cell,

$$\boxed{\frac{d\bar{u}_j}{dt} = -\frac{\hat{f}_{j+\frac{1}{2}} - \hat{f}_{j-\frac{1}{2}}}{\Delta x}}$$

where

$$\hat{f}_{j-\frac{1}{2}}$$

is the numerical flux through the left interface, and

$$\hat{f}_{j+\frac{1}{2}}$$

is the numerical flux through the right interface.

Equivalently,

$$\boxed{\frac{d\bar{u}_j}{dt} = \frac{\text{flux entering} - \text{flux leaving}}{\Delta x}}$$

Thus,

$$\boxed{u = \text{how much exists locally}}$$

and

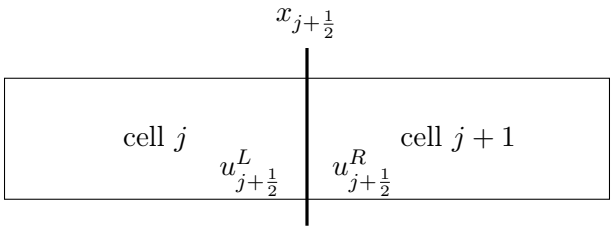
$$\boxed{f(u) = \text{how much is moving through a boundary}}$$

1.4 4. What Is a Numerical Flux?

At an interface $x_{j+\frac{1}{2}}$, WENO or TENO reconstructs two states:

$$u^L_{j+\frac{1}{2}}, \qquad u^R_{j+\frac{1}{2}}.$$

The left value represents the state approaching the interface from cell j, while the right value represents the state approaching from cell $j+1$.



Instead of arbitrarily choosing one of these two states, we define a **numerical flux function**

$$\boxed{\hat{f}_{j+\frac{1}{2}} = \mathcal{F}\left(u^L_{j+\frac{1}{2}}, u^R_{j+\frac{1}{2}}\right)}$$

where $\mathcal{F}$ is a numerical flux or approximate Riemann solver.

The notation

$$\hat{f}$$

indicates that this is a **numerical approximation** of the physical interface flux.

Thus,

$$\boxed{(u^L, u^R) \longrightarrow \text{numerical flux} \longrightarrow \hat{f}_{j+\frac{1}{2}}}$$

The numerical flux estimates how much of the conserved quantity crosses the interface.

1.5 5. Connection Back to WENO/TENO

The complete finite-volume process is:

Step 1 — Cell averages

At time t^n, the solver knows the cell averages

$$\bar{u}^n_j.$$

Step 2 — WENO/TENO reconstruction

Using neighboring cell averages, WENO or TENO reconstructs the left and right interface states:

$$u^L_{j+\frac{1}{2}}, \qquad u^R_{j+\frac{1}{2}}.$$

Step 3 — Numerical flux

The reconstructed states are passed to the numerical flux function:

$$\hat{f}_{j+\frac{1}{2}} = \mathcal{F}\left(u^L_{j+\frac{1}{2}}, u^R_{j+\frac{1}{2}}\right).$$

Step 4 — Determine the rate of change

The flux difference gives the rate of change of the cell average:

$$\boxed{L_j(\bar{u}) = -\frac{\hat{f}_{j+\frac{1}{2}} - \hat{f}_{j-\frac{1}{2}}}{\Delta x}}$$

or

$$\frac{d\bar{u}_j}{dt} = L_j(\bar{u}).$$

Step 5 — Advance in time

A time-integration method such as SSP-RK3 uses this rate of change to advance

$$\boxed{\bar{u}_j^n \longrightarrow \bar{u}_j^{n+1}}$$

where

$$t^{n+1} = t^n + \Delta t.$$

Afterward, the complete process is repeated.

1.6 6. Why Do Fluxes Update the Quantity We Actually Care About?

Consider a control volume containing a conserved quantity such as mass.

Suppose that at time t, the cell contains

$$10 \text{ kg}$$

of material.

During the next second:

- 3 kg enter through the left boundary;
- 1 kg leave through the right boundary.

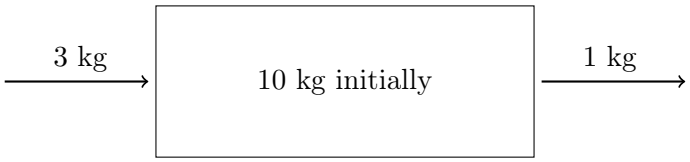
The new amount is therefore

$$10 + 3 - 1 = 12 \text{ kg}.$$

Hence,

$$\boxed{\text{new amount} = \text{old amount} + \text{amount entering} - \text{amount leaving}}$$

This is exactly the physical principle implemented by the finite-volume method.



Because more mass enters than leaves, the mass contained in the cell increases.

If the cell volume remains fixed, its average density also increases.

Thus, the numerical flux is not merely a diagnostic quantity. It directly determines how the conserved state evolves in time.

2 Role of WENO/TENO and RK3

It is useful to separate the spatial and temporal parts of the numerical method.

Spatial discretization

WENO/TENO operate primarily on the spatial structure of the solution:

$$\boxed{\text{WENO/TENO} = \text{spatial reconstruction methods}}$$

They transform neighboring cell averages into accurate left and right interface states.

Together with the numerical flux, they allow us to approximate the spatial operator

$$L(U) = -\frac{\hat{F}_{j+\frac{1}{2}} - \hat{F}_{j-\frac{1}{2}}}{\Delta x}.$$

Temporal discretization

RK3 is responsible for advancing the solution in time:

$$\boxed{\text{RK3} = \text{time-integration method}}$$

Once the spatial discretization gives

$$\frac{dU}{dt} = L(U),$$

RK3 approximately integrates this equation from

$$t^n$$

to

$$t^{n+1} = t^n + \Delta t.$$

3 Why Does RK3 Recompute the Flux?

A simple Forward Euler time step would use

$$U^{n+1} = U^n + \Delta t\, L(U^n).$$

However, the state changes during the time interval.

Therefore, the interface states and fluxes also change.

SSP-RK3 improves the time integration by evaluating the spatial operator several times.

The first stage is

$$U^{(1)} = U^n + \Delta t\, L(U^n).$$

After obtaining $U^{(1)}$, WENO/TENO reconstruction and the numerical flux are recomputed to obtain

$$L(U^{(1)}).$$

The second stage is

$$U^{(2)} = \frac{3}{4}U^n + \frac{1}{4}\left(U^{(1)} + \Delta t\, L(U^{(1)})\right).$$

The reconstruction and fluxes are then recomputed again using $U^{(2)}$.

Finally,

$$U^{n+1} = \frac{1}{3}U^n + \frac{2}{3}\left(U^{(2)} + \Delta t\, L(U^{(2)})\right).$$

Therefore, one RK3 time step can be represented as

$$\boxed{U^n \to L(U^n) \to U^{(1)} \to L(U^{(1)}) \to U^{(2)} \to L(U^{(2)}) \to U^{n+1}}$$

where every evaluation of $L(U)$ requires reconstruction and numerical flux calculation.

4 Complete Numerical Evolution

The full process can be summarized as

State at time t^n

$\downarrow$

Cell averages

$\downarrow$

WENO/TENO reconstruction

$\downarrow$

Left/right interface states

$\downarrow$

Numerical flux

$\downarrow$

Rate of change $L(U)$

$\downarrow$

RK3 time integration

$\downarrow$

State at time t^{n+1}

Then the process is repeated:

$$U^0 \to U^1 \to U^2 \to \cdots \to U^n.$$

5 Key Takeaways

- The quantity u is the physical state we ultimately want to know.
- The flux $f(u)$ describes how that quantity is transported through space.
- The conservation law tells us that the time evolution of u is determined by spatial differences in the flux:

$$u_t = -f(u)_x.$$

- WENO/TENO reconstruct the spatial solution at cell interfaces:

$$\boxed{\text{WENO/TENO} = \text{spatial reconstruction}}$$

- The numerical flux estimates how much conserved quantity crosses each interface:

$$\boxed{\hat{f}_{j+\frac{1}{2}} = \mathcal{F}(u^L, u^R)}$$

- Flux differences determine how much the content of each cell changes:

$$\boxed{\frac{d\bar{u}_j}{dt} = -\frac{\hat{f}_{j+\frac{1}{2}} - \hat{f}_{j-\frac{1}{2}}}{\Delta x}}$$

- RK3 uses these rates of change to advance the solution through time:

$$\boxed{\text{RK3} = \text{time integration}}$$

The complete intuitive description is therefore:

> **WENO/TENO look at the current spatial state and reconstruct the states at the cell boundaries. The numerical flux uses those reconstructed states to determine how conserved quantities are moving between neighboring cells. These fluxes determine how the cell averages are currently changing. RK3 repeatedly reevaluates this evolution at intermediate states and combines those evaluations to obtain an accurate approximation of the state at the next time $t + \Delta t$.**

In compact form,

$$\boxed{\text{state at } t \rightarrow \text{WENO/TENO} \rightarrow \text{interface states} \rightarrow \text{numerical fluxes} \rightarrow \text{rate of change} \rightarrow \text{RK3} \rightarrow \text{state at } t + \Delta t}$$